

Pravan Gorthy

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EDUCATION

University of Texas at Dallas

Bachelor of Science in Computer Science

Richardson, TX

May 2028

TECHNICAL SKILLS

Languages: Java, C++, Python, C#, TypeScript, JavaScript, Bash, SQL, HTML, CSS

Backend/Systems: Distributed Systems, REST APIs, gRPC, Caching, Queues, Observability, Performance Optimization

Frameworks: React, Flask, PyTorch, LangGraph, Node.js, Next.js, Tailwind CSS

Developer Tools: Git, Docker, MCP, XCode, Azure, VS Code, Visual Studio, Figma

EXPERIENCE

Software Engineering Intern

May 2026 – Present

LinkedIn

Mountain View, CA

- Architecting the core **multi-agent orchestration layer** for **Helios** on LinkedIn's internal **DragonSwarm** platform using **LangGraph**, **Python**, and **gRPC**, streamlining request routing and reducing response latency via intent classification, **ReAct** tool-calling agents, and stateful agent-to-agent handoff across specialized agent modes.
- Building the **MCP** tool-integration layer connecting agents to **3 enterprise systems (Workday, SmartRecruiters, ServiceNow)**, designing a reusable schema/auth abstraction that cuts integration time for each new tool connection.
- Designing the **RAG** retrieval and grounding infrastructure backing HR policy answers via optimized chunking/retrieval strategy, and layering in **SSO/SAML** auth and audit logging to meet enterprise security requirements, enabling self-service HR workflows across **Teams, Slack**, and web.

Undergraduate Research Assistant

September 2025 – February 2026

School of Behavioral and Brain Sciences, UT Dallas

Richardson, TX

- Processed large-scale ultrasonic vocalization datasets in **MATLAB** and **Python (VocalMat, Avisoft-SASLab Lite)**, building cleaning and noise-reduction pipelines that improved downstream signal quality for behavioral analysis.
- Generated **publication-ready** visualizations, statistical summaries, and research reports from raw audio recordings, enabling neuroscientists to analyze behavioral patterns and experimental results faster.

Software Engineering Intern

June 2023 – March 2024

HSK Technologies Inc.

Newark, NJ

- Integrated **15+** third-party APIs and built automated **CSV-based ETL pipelines** with scheduling and validation, improving data ingestion, reporting accuracy, and analytics dashboard performance for executive business intelligence.
- Developed **50+ Python automation scripts** for data processing, file handling, validation, and reporting, cutting manual operational workload by an estimated **20+ hours per week**.

PROJECTS

Tab | TypeScript, React, Vite, Express, x402, Base, Anthropic, DexScreener

May 2026

- Built an **agentic payment infrastructure platform** that gives AI agents controlled access to paid tools through budget enforcement, policy-based spend approval, and auditable receipts for every tool call.
- Implemented an **x402**-powered agent spend workflow on **Base** payment rails with real-time policy verdicts, market-data API calls, **Anthropic** inference, and dashboard tracing for approved, denied, and skipped agent actions.

EcoFin Carbon | TypeScript, JavaScript, Tailwind CSS, Gemini, Nessie API, Vite

October 2025

- Built an **AI-powered** fintech web app that analyzes transaction-level spending and estimates carbon emissions using the **Capital One Nessie API**, helping users understand the environmental impact of their purchases.
- Led frontend development across responsive dashboards, user flows, and AI-generated sustainability insights; won **1st Place HackTX 2025** with Capital One recognition and **\$1,000**.

BudgetCents | Python, Flask, React, Stripe, SQL, Git

November 2024

- Built a full-stack personal finance application with **Flask**, **React**, Chart.js dashboards, and **Stripe** payment integration, with SQL-backed expense tracking, budgeting, and transaction history.
- Designed interactive analytics dashboards for budget tracking, category breakdowns, and spending insights; won **1st Place HackFin 2024** and **\$400**.

ACHIEVEMENTS & INTERESTS

- 2x Hackathon Winner:** Won **1st Place HackTX 2025** for EcoFin Carbon and **1st Place HackFin 2024** for BudgetCents, earning over **\$1,400** in prizes across fintech, AI, sustainability, and financial analytics projects.
- BPA:** Top **3** National Finalist in Business and Stock Simulation, applying financial analysis, market research, business strategy, and investment decision-making skills.
- Interests:** FinTech Infrastructure, Backend Engineering, Distributed Systems, Machine Learning, Sustainable Investing, Fitness, Running, Photography